



UA CLINIC SYSTEM

A DIGITAL CLINIC MANAGEMENT PLATFORM FOR HEALTH SERVICES AT THE UNIVERSITY OF THE ASSUMPTION

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ABSTRACT

This study presents the design, development, and evaluation of the UA Clinic System, a web-based platform developed to modernize health service operations at the University of the Assumption. A mixed-methods approach was employed, wherein qualitative data were gathered through surveys and interviews with clinic staff, students, and faculty to identify inefficiencies in manual processes, while quantitative data were collected using the ISO/IEC 25010 software quality standard and the Technology Acceptance Model (TAM). The system was developed using the Agile Prototyping methodology, allowing iterative refinement based on continuous stakeholder feedback. Key features include digital patient registration, secure Electronic Medical Records (EMR) with role-based access control, medical inventory tracking with automated alerts, appointment scheduling, digital medical certificate management, and an AI-powered chatbot (Nursebot). Evaluation results indicate that the system achieved “Very High” ratings across all technical quality and user acceptance metrics, demonstrating high reliability, security, and usability. The study concludes that the UA Clinic System effectively transforms manual clinic processes into a streamlined, data-driven workflow, providing a practical solution for efficient, transparent, and secure campus healthcare management.

Keywords: UA Clinic System, clinic management, Electronic Medical Records (EMR), AI chatbot, inventory monitoring, Agile Prototyping, ISO/IEC 25010, Technology Acceptance Model (TAM), school health services, digital transformation

OBJECTIVES

- ✓ Identify limitations of manual clinic processes
- ✓ Evaluate system quality using ISO/IEC 25010
- ✓ Assess user acceptance using TAM
- ✓ Determine overall system effectiveness

METHODS

- 📋 Planning & Analysis: Defining the scope and core user requirements.
- 📐 Design & Prototyping: Iterative building of the MERN-stack architecture.
- ⚙️ Testing & Evaluation: Technical assessment and user acceptance gathering.
- ✅ Deployment: Official implementation at the UA Clinic.

CONCLUSION

The UA Clinic System effectively addressed the major operational challenges of the University of the Assumption clinic by replacing manual, paper-based processes with a centralized digital platform. The system improved record management, inventory monitoring, appointment scheduling, and data security through role-based access and verified institutional accounts. Evaluation results using ISO/IEC 25010 and the Technology Acceptance Model revealed very high ratings in security, usability, performance, and user acceptance. These findings confirm that the system is technically reliable, user-friendly, and well-aligned with the needs of clinic staff, students, and faculty. Overall, the UA Clinic System serves as a practical and sustainable digital solution for enhancing campus healthcare service delivery.

RESULTS

The evaluation results indicate that the UA Clinic System achieved "Very High" ratings across all technical and user metrics. IT experts gave the system a grand mean of 3.44 for technical quality, with Security receiving the highest rating of 3.73, highlighting its robust data protection. Furthermore, end-users reported high satisfaction with a TAM grand mean of 3.60, specifically praising the system's Ease of Use (3.61) and its ability to complete clinic tasks faster than manual methods. These scores validate that the system is a reliable, high-performance tool ready for university-wide adoption.

